

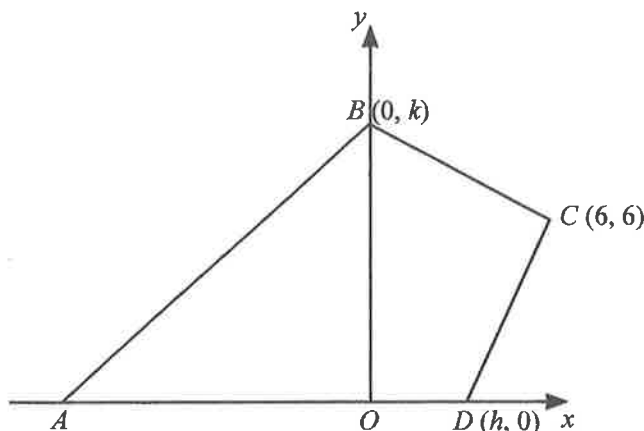
Basic

1. The equation of a curve is $y = 2x^2 - 8x + 11$.
- (a) By expressing $2x^2 - 8x + 11$ in the form $a(x + b)^2 + c$, where a , b and c are constants, find the coordinates of the stationary point on the curve. [2]
- (b) The line $y = 2x + 3$ intersects the curve at points A and B . Find the value of k for which the distance AB can be expressed as $\sqrt{k}$. [4]
- [N22/I/1]
2. The equation of a circle is $x^2 + y^2 + 4x - 6y - 12 = 0$.
- (i) Find the radius of the circle and the coordinates of its centre. [4]
- (ii) Find the equation of the tangent to the circle at the point $(1, 7)$. [3]
- [N20/I/9]

Intermediate

3. The triangle ABC is such that A is $(4, 2)$, B is $(10, 2)$ and C is (p, q) where $q > 0$. The area of triangle ABC is 15 units² and $AC = \sqrt{89}$ units.
- (a) Explain why $q = 7$. [3]
- (b) Find the possible values of p . [4]
- [N22/I/11]

4.

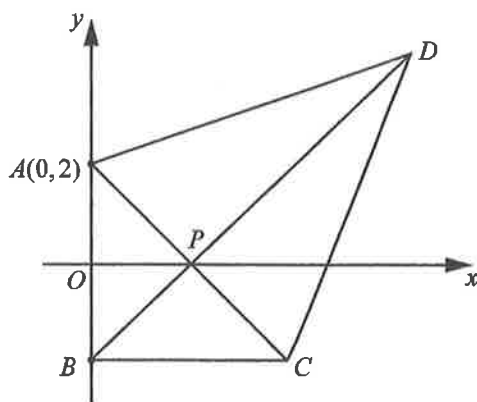


The diagram shows a kite $ABCD$ in which $AB = AD$ and $BC = CD$. Points A and D lie on the x -axis and B lies on the y -axis. Point C is $(6, 6)$, D is $(h, 0)$ and B is $(0, k)$, where h and k are positive constants and $k \neq h$.

- (a) Show that $h + k = 12$. [3]
- It is now given that $h = 4$.
- (b) Find the coordinates of A . [4]
- (c) Find the area of the kite. [2]

[N21/I/9]

5.



The diagram shows a kite $ABCD$ in which $AB = BC$ and $AD = DC$. The points $A(0, 2)$ and B lie on the y -axis. The diagonals AC and BD intersect at the point P on the x -axis.

Given that the length of AB is 4 units,

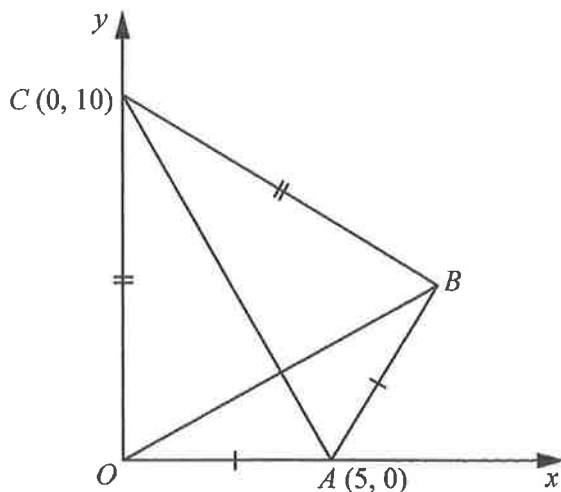
- (i) explain why BC is parallel to the x -axis, [2]
- (ii) find the coordinates of C . [1]

Given further that the area of the kite is 28 units^2 ,

- (iii) find the coordinates of D . [5]

[N19/I/9]

6.



The diagram shows a kite $OABC$ in which $OA = AB$ and $OC = BC$. Given that the coordinates of A and C are $(5, 0)$ and $(0, 10)$ respectively, find

- (i) the equation of OB , [3]
- (ii) the coordinates of B . [4]

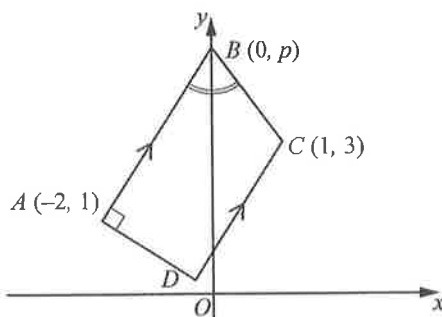
[N18/I/9]

7. Three points are given by $A(1, 4)$, $B(9, 8)$ and $C(7, 12)$.

- (i) Show that angle ABC is 90° . [3]
- (ii) Explain why A , B and C lie on a circle with diameter AC . [1]
- (iii) Find the equation of the circle. [3]
- (iv) Explain why the tangent to the circle at B is parallel to the y -axis. [2]
- (v) Find the equation of the tangent to the circle at C . [2]

[N18/II/11]

8.

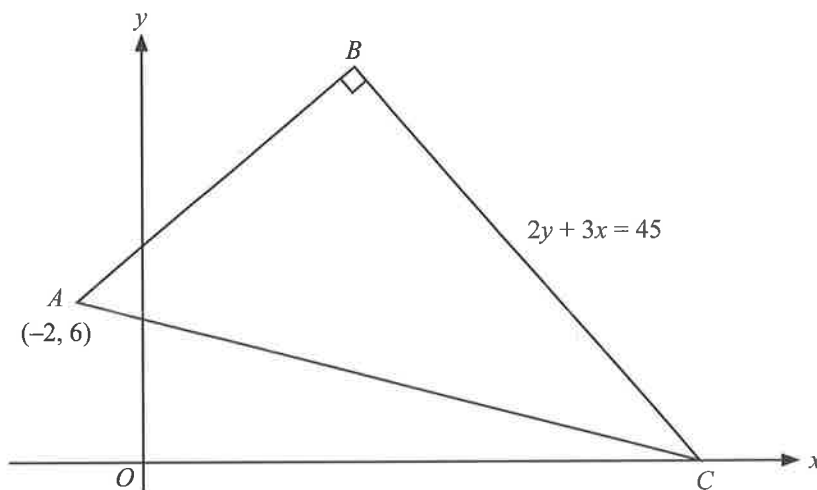


The diagram shows a trapezium with vertices $A(-2, 1)$, $B(0, p)$, $C(1, 3)$ and D . The sides AB and DC are parallel and the angle DAB is 90° . Angle ABO is equal to angle CBO .

- (i) Express the gradients of the lines AB and CB in terms of p and hence, or otherwise, show that $p = 5$. [4]
- (ii) Find the coordinates of the point D . [6]
- (iii) Find the area of the trapezium $ABCD$. [2]

[N17/II/9]

9.



The diagram shows a triangle ABC in which the point A is $(-2, 6)$, the point C lies on the x -axis and angle ABC is 90° . The equation of BC is $2y + 3x = 45$.

- (i) Find the coordinates of B . [5]
- (ii) Given that M is the midpoint of AC and that $ABCD$ is a rectangle, find the coordinates of M and of D . [3]

[N16/I/8]

10. The equation of a circle, C_1 , with centre A , is $x^2 + y^2 - 4x - 2y = 95$.

- (i) Find the coordinates of A and the radius of C_1 . [4]
 - (ii) Show that the point $P(10, 7)$ lies on C_1 . [1]
 - (iii) Find the equation of the tangent to C_1 at P . [3]
- A second circle, C_2 , has diameter AP .
- (iv) Find the equation of C_2 . [3]
 - (v) Find the equation of the tangent to C_2 at P . [1]

[N16/II/11]

11. The equation of a circle is $x^2 + y^2 - 16x - 2y + 40 = 0$.

- (a) Find the radius and coordinates of the centre of the circle. [4]
 (b) Find the shortest distance of the centre of the circle from the line $y = 2x$ and hence explain why the circle does not intersect the line $y = 2x$. [5]

The point $C(11, 5)$ lies on the circle and CD is a diameter of the circle.

- (c) Find the coordinates of D . [2]

[N22/II/9]

12. The equation of a circle is $(x + 2a)^2 + (y - a)^2 = ka^2$, where a and k are positive constants.

It is given that $k = 4$.

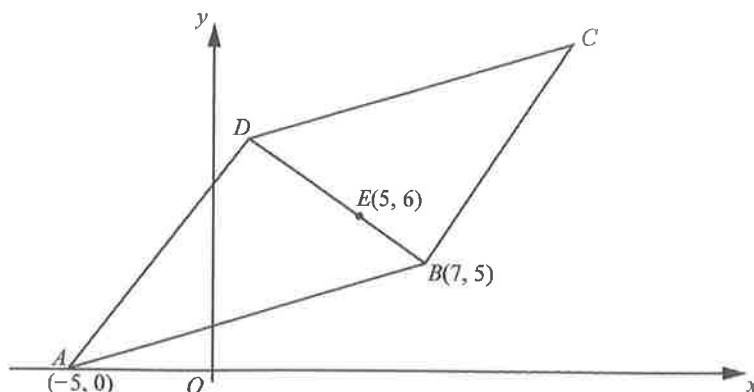
- (a) Explain why the y -axis is a tangent to the circle. [3]
 (b) Find, in terms of a , the coordinates of the points on the circle at which the tangent to the circle is parallel to the x -axis. [2]

It is now given that $k = 5$.

- (c) Verify that the circle passes through the origin O . [1]
 (d) Given that OP is a diameter of the circle, find, in terms of a , the coordinates of the point at which the tangent to the circle at P meets the x -axis. [6]

[N21/II/9]

13.



The diagram shows a parallelogram with vertices $A(-5, 0)$, $B(7, 5)$, C and D . The point $E(5, 6)$ lies on the diagonal BD of the parallelogram. The perimeter of the parallelogram is 46 units.

- (i) Show that the x -coordinate of D satisfies the equation $5x^2 + 6x = 11$. [7]
 (ii) Determine the coordinates of D , explaining why the diagram is necessary. [3]
 (iii) Find the coordinates of C . [1]

[N20/II/9]

14. A tangent to a circle at the point $(5, 5)$ passes through the origin. The line with equation $3y = x + 2$ is a normal to the circle.

- (i) Showing all your working, find the equation of the circle. [7]
 (ii) Find the coordinates of the point on the circle which is nearest to the x -axis. [2]

[N19/II/6]

15. The equation of the normal to the circle $x^2 + y^2 + 8x - 24y + 96 = 0$ at the point R is $3y + 4x = k$.

- (i) Find the value of the constant k . [5]

The normal to the circle at R meets the x -axis at the point S .

- (ii) Given that R lies between S and the centre of the circle, find the length of RS . [5]

[N17/I/12]

16. The positive x - and y -axes are tangents to a circle C .

(i) What can be deduced about the coordinates of the centre of C ?

[1]

The line T is a tangent to C at the point $(9, 8)$ on the circle. Given that the centre of C lies below and to the left of $(9, 8)$, find

(ii) the equation of C ,

[5]

(iii) the equation of T .

[3]

[N15/II/7]

